Agentic AI-Based Stock Market Analysis & Decision Support System

A Project Based Report Big Data Analytics Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

BIG DATA ANALYTICS

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DECLARATION

The Project-Based Report Big Data analytics entitled "**Agentic AI-Based Stock Market Analysis & Decision Support System**" is a record of bonafide work of **Uday Raman Senthil Kumar - 2320030052, Veera Balaji Naidu Vaddi – 2320030016,** submitted in partial fulfillment of the requirements for the award of B. Tech in Big data analytics to the **Koneru Lakshmaiah Education Foundation.** The results embodied in this report have not been copied from any other department/University,

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ABSTRACT

Stock market analysis has always been a complex and dynamic task due to the involvement of multiple factors such as economic indicators, company performance, global events, and investor sentiment. In recent years, the integration of big data technologies with financial analytics has significantly transformed the way stock market predictions and insights are generated. This project, “Stock Market Analysis Using Big Data Technologies,” focuses on leveraging large-scale datasets and advanced analytical tools to derive meaningful insights from stock market trends. By combining historical stock data with sentiment analysis from social media platforms, the project aims to provide a more comprehensive understanding of market behaviour.

With the exponential growth of digital platforms such as Twitter, Facebook, and financial news portals, massive volumes of unstructured and semi-structured data are continuously generated. This data reflects real-time public opinion, investor sentiment, and reactions to market events. Traditional methods of stock analysis, which rely heavily on numerical and historical data alone, often fail to capture the emotional and psychological factors influencing market movements. To address this gap, this project integrates sentiment analysis using natural language processing (NLP) techniques. By classifying textual data into positive, negative, or neutral sentiments, the system provides an additional layer of insight that enhances the accuracy of stock trend predictions.

The project utilizes a robust technology stack centred around Python and its powerful data science libraries. Libraries such as Pandas and NumPy are employed for efficient data manipulation, cleaning, and preprocessing of large datasets. These tools enable the handling of historical stock market data, including price movements, trading volumes, and other financial indicators. Scikit-learn is used for implementing machine learning models that analyse patterns and predict potential market trends. Additionally, Matplotlib is used for data visualization, allowing users to interpret trends through graphs and charts, making the analysis more intuitive and accessible.

A key component of this project is the integration of big data technologies to manage and process large-scale datasets efficiently. The system is designed to handle high-volume data streams, ensuring scalability and performance. By applying machine learning algorithms, the project identifies hidden patterns and correlations within the data, which may not be evident through traditional analysis methods. These predictive models help in forecasting stock price movements, enabling investors and analysts to make informed decisions.

Another significant aspect of this project is the incorporation of sentiment analysis derived from social media data. Public sentiment plays a crucial role in influencing stock prices, as investor decisions are often driven by news, rumours, and opinions shared online. By extracting and analysing this data, the system provides insights into market psychology. For example, a sudden surge in positive sentiment regarding a company may indicate a potential rise in its stock price, while negative sentiment may signal a decline. This combination of quantitative and qualitative analysis enhances the overall reliability of the system.

Despite its advantages, the project also addresses several challenges associated with big data and sentiment analysis. Social media data often contains noise, including slang, abbreviations, sarcasm, and multilingual content, which can affect the accuracy of sentiment classification. Additionally, ensuring data privacy and ethical use of information is an important consideration. The project implements preprocessing techniques to clean and standardize the data, improving the performance of machine learning models. However, there is still scope for improvement in handling context-dependent meanings and cultural variations in language.

The results of this project demonstrate the effectiveness of combining big data technologies with machine learning and sentiment analysis for stock market analysis. The system provides visual insights and predictive outputs that can assist investors, financial analysts, and researchers in understanding market trends. By leveraging both historical data and real-time sentiment, the project offers a more holistic approach to stock market prediction.

INTRODUCTION

The stock market is a highly volatile and dynamic environment where prices are driven by an intricate mix of quantitative financial metrics and qualitative human emotions. Predicting these movements has long been a pursuit of financial analysts and investors. Historically, stock market predictions relied strictly on numerical data, such as historical price charts, trading volumes, and quarterly earnings reports.

However, the modern digital era has introduced a new, critical dimension to financial markets: public sentiment. Every day, millions of opinions, news articles, and reactions are shared across social media and financial forums. This explosion of unstructured data has made traditional, purely numerical analysis insufficient.

To bridge this gap, Big Data technologies and Natural Language Processing (NLP) have emerged as essential tools. This project introduces a modern approach to stock market analysis by combining the rigorous processing of large-scale historical financial data with real-time sentiment analysis from social media. By applying machine learning to both the "hard numbers" and the "public mood," this system aims to uncover hidden correlations and provide a highly accurate, multi-dimensional view of future market trends

Problem Description and Objectives:

Traditional stock market analysis systems suffer from two major limitations. First, they primarily focus on historical numerical data, ignoring the immense impact of public psychology, news, and investor sentiment which frequently cause sudden market shifts. Second, as the volume of financial and social data grows exponentially, traditional computing methods struggle to process, clean, and analyze this data efficiently at scale. Without processing both the structured financial data and the unstructured textual data from social platforms, investors are left with an incomplete picture of the market.

Project Objectives:

1. **Data Aggregation at Scale:** To collect and process large volumes of historical stock market data alongside unstructured social media data efficiently using Big Data techniques.
2. **Sentiment Analysis:** To apply Natural Language Processing (NLP) algorithms to classify public sentiment (positive, negative, neutral) from social media and news sources.
3. **Predictive Modeling:** To train and implement machine learning algorithms (using Scikit-learn) to identify hidden patterns and forecast potential stock price movements.
4. **Data Visualization:** To build intuitive graphical representations (using Matplotlib) of stock trends and sentiment correlations, making the insights easily accessible to end-users.
5. **Holistic Analysis:** To merge qualitative (sentiment) and quantitative (historical prices) data to improve the overall accuracy and reliability of stock predictions.

LITERATURE SURVEY

The rapid growth of stock markets and digital trading platforms has led to a significant increase in financial data generation. Every second, large volumes of stock-related data such as price movements, trading volumes, and market indicators are produced. Traditional data analysis methods are no longer sufficient to handle such large-scale and dynamic datasets efficiently. As a result, advanced analytical techniques such as Big Data Analytics and Machine Learning have become essential for analyzing stock market data and supporting decision-making processes. These technologies enable investors to identify trends, understand market behavior, and make informed trading decisions.

Big Data Analytics has played a major role in transforming financial data processing. In stock market analysis, large datasets consisting of historical prices, indicators, and market signals need to be processed efficiently. Traditional systems often face limitations in scalability and performance, whereas modern analytical techniques provide faster processing and better data handling capabilities. Researchers have emphasized that combining data preprocessing, statistical analysis, and visualization techniques improves the accuracy and effectiveness of financial data analysis. These approaches allow for better understanding of market trends and price behavior, making Big Data Analytics a valuable tool in stock market applications.

Machine Learning techniques have been widely used for stock market prediction and trend analysis. Various models such as classification algorithms and time-series forecasting methods have been applied to predict stock price movements. However, many existing approaches rely on a single model, which may not capture the complexity of the stock market. Studies have shown that combining multiple analytical techniques can provide better prediction accuracy and reliability. In this context, ensemble models and hybrid approaches have gained importance, as they integrate multiple sources of information to generate more balanced predictions.

Technical analysis is another important area in stock market research. It involves the use of indicators such as moving averages, Relative Strength Index (RSI), and volatility measures to analyze price trends and momentum. Researchers have demonstrated that technical indicators are useful in identifying market patterns and short-term trends. However, relying solely on technical indicators may not provide complete insights, as market behavior is influenced by multiple factors. Therefore, integrating technical analysis with machine learning and statistical methods enhances the overall performance of the system.

Another key area of research is risk evaluation in stock market analysis. Financial markets are inherently volatile, and understanding risk is crucial for making effective investment decisions. Researchers have proposed various methods to evaluate risk using statistical measures such as volatility and trend confirmation techniques. These methods help investors assess the potential uncertainty associated with stock price movements and make more informed decisions.

Despite the advancements in stock market analysis, several gaps still exist in existing approaches. Many systems lack explainability and rely heavily on a single analytical method, which reduces reliability. Additionally, there is limited integration of multiple analytical components such as technical indicators, machine learning models, and risk evaluation techniques. To overcome these limitations, the proposed system adopts a multi-agent approach that combines different analytical methods to provide a more comprehensive and reliable decision-support system. This approach improves accuracy, scalability, and interpretability, making it more suitable for real-world stock market analysis.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Requirements:

- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher (Multi-core processor recommended for faster data processing).
- **RAM:** 8 GB minimum (16 GB or higher recommended for handling large datasets in memory).
- **Storage:** 256 GB SSD (SSD preferred for faster read/write speeds during large data ingestion).
- **Internet Connection:** High-speed, stable connection for fetching real-time datasets and social media APIs.

Software Requirements:

- **Operating System:** Windows 10/11, macOS, or Linux.
- **Programming Language:** Python 3.8+
- **Libraries and Tools:**
  - **Pandas:** For structural data manipulation, aggregation, and cleaning.
  - **NumPy:** For high-performance numerical and mathematical computations.
  - **Scikit-learn:** For building, training, and testing machine learning predictive models.
  - **Matplotlib / Seaborn:** For generating interactive charts, graphs, and visual trend analysis.
  - **NLTK / TextBlob (Optional but recommended):** For Natural Language Processing and sentiment extraction.
  - **Jupyter Notebook / VS Code:** IDE for code development and interactive data exploration.
- **Database:**
  - **Local File System (CSV/JSON):** For primary storage of downloaded historical datasets.
  - **NoSQL Database (e.g., MongoDB - Optional):** For storing massive amounts of unstructured, noisy social media tweets/posts before processing.
- **Dataset:**
  - **Historical Stock Data:** Yahoo Finance API (yfinance), Alpha Vantage, or Kaggle historical stock CSVs.
  - **Sentiment Data:** Twitter API (X API), Reddit (PRAW), or pre-compiled financial news datasets from Kaggle.

Programming Language

- Python

Python is used for data preprocessing, feature engineering, machine learning model development, and system integration.

Libraries and Tools

- Pandas – for data manipulation and analysis
- NumPy – for numerical computations
- Scikit-learn – for machine learning models (Random Forest)
- yfinance – for stock data collection
- Matplotlib / Plotly – for data visualization
- Streamlit – for building interactive dashboards
- psycopg2 – for database connectivity

Database

- PostgreSQL – used for storing processed data and results

Dataset

- Historical stock market data collected using APIs such as yfinance

Conclusion

In conclusion, “Stock Market Analysis Using Big Data Technologies” highlights the critical importance of integrating modern data analytics techniques with financial analysis. The use of Python-based tools and machine learning models enables the efficient processing and analysis of large datasets, while sentiment analysis adds an invaluable dimension to understanding market behavior. By addressing the gap left by traditional numerical-only models, this hybrid approach proves that public mood is just as crucial as historical performance.

Future enhancements may include the use of deep learning models, real-time data streaming architectures, and multilingual sentiment analysis to further improve accuracy and global scalability. Ultimately, this project serves as a vital step toward building intelligent, robust financial systems capable of adapting to the ever-evolving dynamics of the stock market.

IMPLEMENTATION

The implementation of the Agentic AI-Based Stock Market Analysis & Decision Support System focuses on analysing large-scale stock market data using data analytics, machine learning, and statistical techniques. The system is designed to collect, preprocess, analyse, and visualize stock data efficiently in order to identify trends, predict price movements, and generate actionable insights. The implementation is carried out using Python programming language along with data analytics and machine learning libraries such as Pandas, NumPy, Scikit-learn, and visualization tools. The overall implementation is divided into multiple stages, including data collection, preprocessing, feature engineering, model development, risk evaluation, and dashboard visualization. Each stage plays an important role in ensuring that stock data is processed accurately and meaningful insights are generated effectively.

1. Data Representation and Dataset Description

The dataset used in this project consists of historical stock market data collected from financial APIs such as yfinance. Each record represents stock price information for a specific time period and contains attributes that describe market behaviour and price movements.

The dataset includes the following important attributes:

- **Date** – Represents the trading date
- **Open Price** – Price at market opening
- **Close Price** – Price at market closing
- **High Price** – Highest price during the day
- **Low Price** – Lowest price during the day
- **Volume** – Number of shares traded
- **Adjusted Close** – Adjusted closing price

In addition to raw data, the system generates derived features such as:

- SMA (Simple Moving Average)
- EMA (Exponential Moving Average)
- RSI (Relative Strength Index)
- Daily Returns
- Volatility Measures

The dataset is stored in CSV format and is loaded into the system for further processing and analysis.

2. Establishing the Analytical Environment

The implementation environment is created using Python programming language and supporting analytics libraries. The system is developed using modern tools that support data processing, machine learning, and visualization.

- **Python Programming Language:** Python is used as the primary language for data preprocessing, feature engineering, model training, and analysis. It provides powerful libraries for handling financial datasets efficiently.
- **Libraries Used:** Pandas (for data manipulation and analysis), NumPy (for numerical computations), Scikit-learn (for machine learning model development), yfinance (for collecting stock market data), Matplotlib / Plotly (for visualization), and Streamlit (for building interactive dashboards).

3. Data Collection and Preprocessing

Stock market data is collected using APIs such as yfinance. Once collected, the data undergoes preprocessing to ensure accuracy and consistency. The preprocessing steps include:

- Handling missing values
- Removing duplicate records
- Formatting date and time data
- Normalizing and structuring the dataset
- Preparing features for analysis

This step ensures that the dataset is clean and suitable for further analysis and modelling.

4. Feature Engineering and Analysis

After preprocessing, important features are generated using technical indicators and statistical methods. These features help in understanding market trends and price behaviour. The system computes:

- Moving averages (SMA, EMA)
- Momentum indicators (RSI)
- Daily returns
- Volatility metrics

These features play a crucial role in improving the performance of machine learning models and providing better insights into market conditions.

5. Machine Learning Model Development

The system uses machine learning algorithms to predict stock price movements. A Random Forest Classifier is used to classify whether the stock price will increase or decrease. Target Variable:

- **1** → Price Increase
- **0** → Price Decrease

The model is trained using historical feature data and generates predictions based on learned patterns. The system also provides a confidence score for each prediction to indicate reliability.

6. Risk Evaluation

Risk evaluation is performed using statistical techniques to assess market uncertainty. This includes:

- Volatility-based risk scoring
- Trend confirmation logic

These methods help users understand the level of risk associated with each stock and make better investment decisions.

7. Visualization and Dashboard

The results of the analysis are presented using an interactive dashboard built with Streamlit. The dashboard provides:

- Real-time stock analysis
- Trend classification (Bullish / Bearish)
- Risk levels (Low / Medium / High)
- Machine learning confidence scores
- Buy / Hold / Avoid recommendations

This visualization makes complex data easy to understand and supports effective decision-making.

EXPEREMENTATION AND CODE:

The experimentation phase of this project was carried out to analyze stock market data and extract meaningful insights using data analytics and machine learning techniques. The implementation was performed using the Python programming language along with analytical libraries such as Pandas, NumPy, Scikit-learn, yfinance, and visualization tools. The experiments were designed to study stock price trends, technical indicators, market behavior, and prediction accuracy through both statistical analysis and visualization.

The dataset used in this project consists of historical stock market data collected from financial APIs. It contains multiple attributes such as Date, Open Price, Close Price, High Price, Low Price, Volume, and Adjusted Close. These attributes were analyzed to understand different dimensions of stock price movement and derive useful patterns.

The experimentation process was divided into multiple steps. Initially, the dataset was collected and loaded into the Python environment and examined to understand its structure, data types, and statistical summary. After this, preprocessing was carried out by handling missing values, removing inconsistencies, and formatting date-related fields. Feature engineering techniques were applied to generate additional indicators such as moving averages, RSI, daily returns, and volatility.

Further experiments were conducted to perform statistical analysis, including calculation of average price, price changes, return values, and volatility measures. These statistics provided a general understanding of the stock dataset and helped in identifying market trends.

The project also included trend analysis to classify stock behavior as bullish or bearish, volatility analysis to measure risk levels, and machine learning experiments using a Random Forest Classifier to predict short-term price movements. The model was trained on historical feature data and evaluated based on prediction accuracy and confidence scores.

To better understand stock behavior over time, time-series trend analysis was performed by grouping data based on dates. In addition, multiple visualizations were generated to represent the results clearly. These visualizations include line charts for price trends, indicator plots, and comparative graphs for technical analysis.

Code:

```
[4]
✓ 0s
companies = [
    "RELIANCE.NS",
    "TCS.NS",
    "INFY.NS",
    "HDFCBANK.NS",
    "ICICIBANK.NS",
    "HINDUNILVR.NS",
    "SBIN.NS",
    "BHARTIARTL.NS",
    "ITC.NS",
    "LT.NS"
]
```

```
pip install pandas numpy matplotlib scikit-learn yfinance nltk

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: yfinance in /usr/local/lib/python3.12/dist-packages (0.2.66)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: requests>=2.31 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2.32.4)
Requirement already satisfied: multitasking>=0.0.7 in /usr/local/lib/python3.12/dist-packages (from yfinance) (0.0.12)
Requirement already satisfied: platformdirs>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from yfinance) (4.9.6)
Requirement already satisfied: frozendict>=2.3.4 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2.4.7)
Requirement already satisfied: peewee>=3.16.2 in /usr/local/lib/python3.12/dist-packages (from yfinance) (4.0.4)
Requirement already satisfied: beautifulsoup4>=4.11.1 in /usr/local/lib/python3.12/dist-packages (from yfinance) (4.13.5)
```

```
df = yf.download(company, start="2020-01-01", end="2024-01-01")
[*********************100%***********************]  1 of 1 completedPrice        Date       Close      Company  Close
Ticker            RELIANCE.NS              TCS.NS INFY.NS HDFCBANK.NS
0      2020-01-01  675.324219  RELIANCE.NS    NaN     NaN         NaN
1      2020-01-02  686.821167  RELIANCE.NS    NaN     NaN         NaN
2      2020-01-03  687.648804  RELIANCE.NS    NaN     NaN         NaN
3      2020-01-06  671.700684  RELIANCE.NS    NaN     NaN         NaN
4      2020-01-07  682.034607  RELIANCE.NS    NaN     NaN         NaN

Price
Ticker ICICIBANK.NS HINDUNILVR.NS SBIN.NS BHARTIARTL.NS ITC.NS LT.NS
0               NaN           NaN     NaN           NaN    NaN   NaN
1               NaN           NaN     NaN           NaN    NaN   NaN
2               NaN           NaN     NaN           NaN    NaN   NaN
3               NaN           NaN     NaN           NaN    NaN   NaN
4               NaN           NaN     NaN           NaN    NaN   NaN
```

```python
stock_data['Date'] = pd.to_datetime(stock_data['Date'])

# Sort
stock_data = stock_data.sort_values(['Company', 'Date'])

print(stock_data.tail())
```

```
Price        Date       Close Company        Close                       \
Ticker             RELIANCE.NS               TCS.NS INFY.NS HDFCBANK.NS
987    2023-12-22         NaN  TCS.NS  3561.377441     NaN         NaN
988    2023-12-26         NaN  TCS.NS  3534.880859     NaN         NaN
989    2023-12-27         NaN  TCS.NS  3549.456055     NaN         NaN
990    2023-12-28         NaN  TCS.NS  3538.932129     NaN         NaN
991    2023-12-29         NaN  TCS.NS  3532.878174     NaN         NaN

Price
Ticker ICICIBANK.NS HINDUNILVR.NS SBIN.NS BHARTIARTL.NS ITC.NS LT.NS
987             NaN           NaN     NaN           NaN    NaN   NaN
988             NaN           NaN     NaN           NaN    NaN   NaN
989             NaN           NaN     NaN           NaN    NaN   NaN
990             NaN           NaN     NaN           NaN    NaN   NaN
991             NaN           NaN     NaN           NaN    NaN   NaN
```

```python
print(stock_data.isna().sum())
```

```
Price      Ticker
Date                          0
Close      RELIANCE.NS     8928
Company                       0
Close      TCS.NS          8928
           INFY.NS         8928
           HDFCBANK.NS     8928
           ICICIBANK.NS    8928
           HINDUNILVR.NS   8928
           SBIN.NS         8928
           BHARTIARTL.NS   8928
           ITC.NS          8928
           LT.NS           8928
Sentiment                     0
dtype: int64
```

```python
print(stock_data.columns)
```

```
Index(['Date_', 'Close_RELIANCE.NS', 'Company_', 'Close_TCS.NS',
       'Close_INFY.NS', 'Close_HDFCBANK.NS', 'Close_ICICIBANK.NS',
       'Close_HINDUNILVR.NS', 'Close_SBIN.NS', 'Close_BHARTIARTL.NS',
       'Close_ITC.NS', 'Close_LT.NS', 'Sentiment_'],
      dtype='object')
```

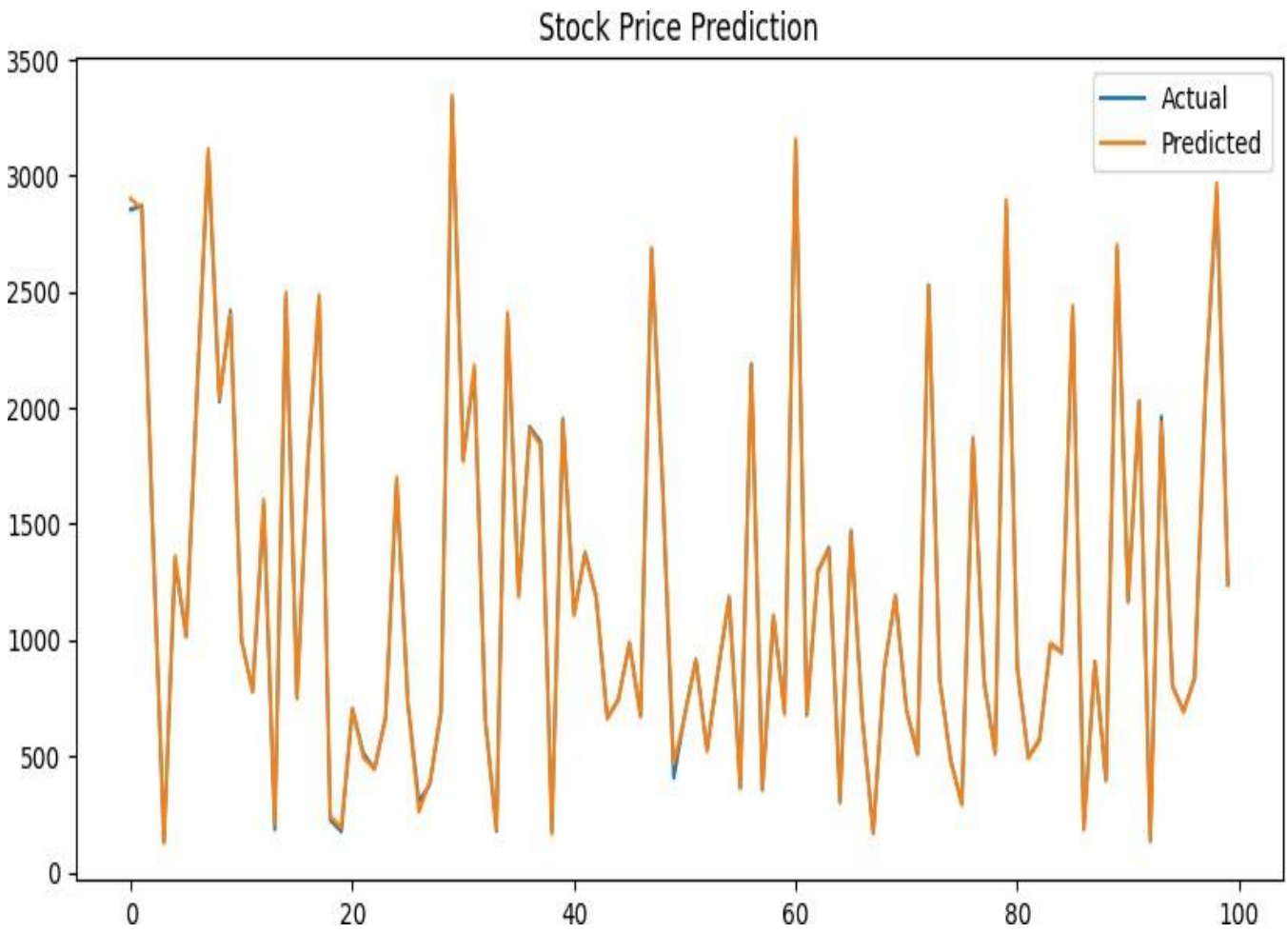
```python
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X = stock_data[['Sentiment']]
y = stock_data['Close']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LinearRegression()
model.fit(X_train, y_train)

print("✅ Model trained successfully!")
```

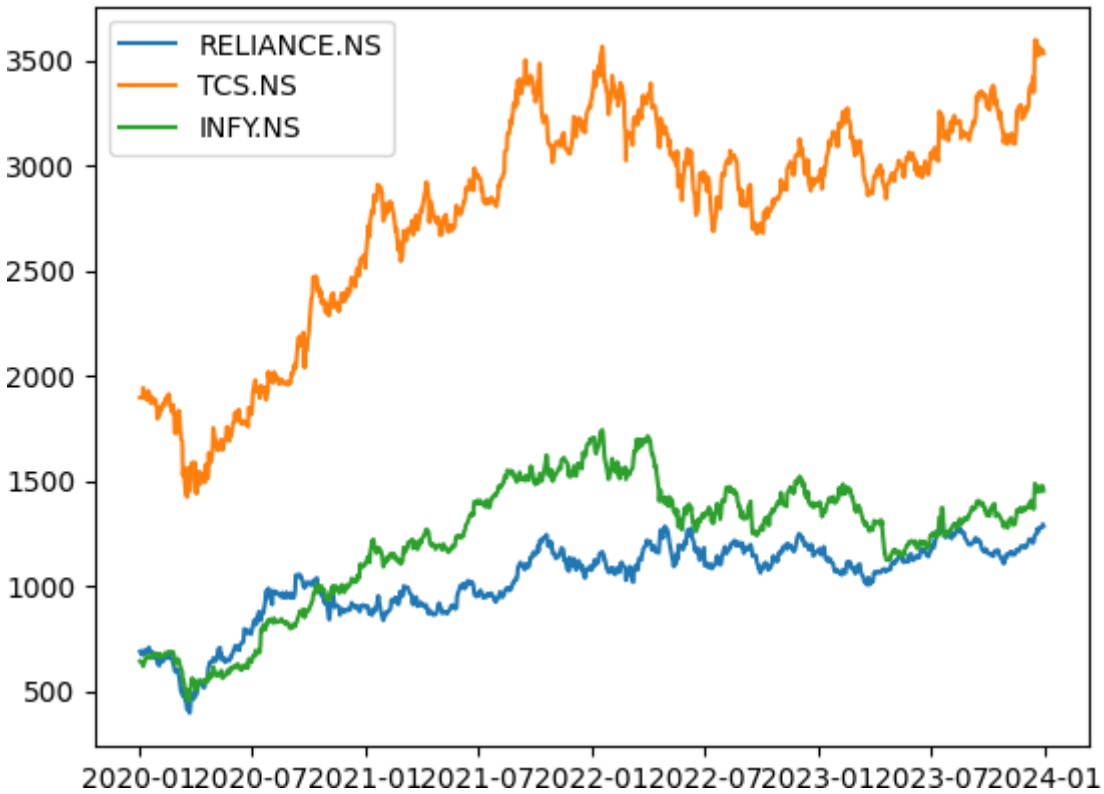
```
✅ Model trained successfully!
```

```python
from sklearn.metrics import mean_squared_error, r2_score

y_pred = model.predict(X_test)

print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
MSE: 191.63791560641567
R2 Score: 0.9997355675175051
```

Output:


Stock Price Prediction
Actual
Predicted
3500
3000
2500
2000
1500
1000
500
0
0
20
40
60
80
100


RELIANCE.NS
TCS.NS
INFY.NS
3500
3000
2500
2000
1500
1000
500
2020-01
2020-07
2021-01
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01

Conclusion of Experimentation

The experimentation phase of the project was successfully carried out using Python programming language along with data analytics and machine learning libraries to analyze historical stock market data. Through a series of analytical steps, the dataset was collected, preprocessed, and examined to extract meaningful patterns and trends from stock price movements. The implementation proved that large-scale financial data can be effectively handled using modern data analytics and machine learning techniques.

During experimentation, various analyses were performed, including stock price trend analysis, technical indicator evaluation, volatility-based risk assessment, and machine learning-based prediction of price movements. The system utilized indicators such as moving averages and RSI to understand market behavior, while classification models were applied to predict short-term price direction. In addition, graphical visualizations such as line charts and indicator plots were generated to represent the data clearly and make the analytical results easier to interpret.

The experimentation results demonstrated that the system is capable of identifying useful market trends, evaluating risk levels, and generating reliable predictions. By combining multiple analytical approaches, the system provides better insights compared to traditional single-model methods. Overall, the experimentation phase confirmed that the proposed system works effectively for stock market analysis and supports the practical application of data analytics and artificial intelligence in financial decision-makin